

Product Optimization & Revenue Contribution Analysis

A Research Paper on Afficionado Coffee Roasters' Product & Sales Data

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Abstract

This paper presents a product-level analysis of sales data from Afficionado Coffee Roasters, covering 149,116 transactions across three New York stores (Lower Manhattan, Hell's Kitchen, and Astoria) over 2025. The goal was to understand which products actually drive the business - by revenue, by volume, and by profitability per unit - and to translate that into practical recommendations for menu optimization. The analysis follows a full workflow: data cleaning in Python, exploratory analysis in MySQL, and visual analysis in Power BI and Streamlit. The key finding is that the business is heavily concentrated around a small set of products - just 11 of 29 product types account for 80% of total revenue - which has direct implications for inventory priority, pricing, and menu planning.

1. Introduction

Afficionado Coffee Roasters operates across three store locations and sells a wide range of products - not just brewed coffee, but tea, bakery items, packaged beans, drinking chocolate, and a handful of branded merchandise items. With 80 individual SKUs spread across 9 product categories, it isn't obvious at a glance which products are actually carrying the business and which ones are sitting on the menu without contributing much. This project set out to answer that question using the store's own transaction data.

The analysis was carried out as part of a Business Analyst internship, with the objective of practicing the full analytics pipeline - from raw, messy transaction data to a decision-ready dashboard - while producing insights that a real coffee business could act on.

2. Objectives

- Identify which products and categories generate the most revenue and the most sales volume.
- Understand how concentrated the business's revenue is around a small number of "hero" products.
- Flag underperforming products that may be candidates for menu review.
- Translate these findings into clear, actionable recommendations for menu and inventory decisions.

3. Data & Methodology

3.1 Data Source

The dataset consists of 149,116 individual sales transactions recorded across three store locations during 2025. Each row captures a transaction ID, date and time, store location, product ID, unit price, product category, product type, product detail, quantity sold, and the resulting revenue.

3.2 Tools & Workflow

- **Python (pandas), in Jupyter Notebook:** used to clean the raw dataset - handling missing values, correcting data types, and preparing a clean CSV for downstream analysis.
- **MySQL:** used for exploratory data analysis, including ranking, Pareto, and revenue-contribution queries.
- **Power BI:** used to build a three-page interactive report covering business overview, product performance, and revenue concentration.
- **Streamlit:** used to rebuild the same analysis as a live, filterable web application, with dynamic Top-N and category filters.

3.3 Scope & Limitations

The dataset covers a single calendar year (2025) and does not include cost or margin data, so this analysis is based on revenue and volume rather than profitability. It also does not include customer-level data, so findings are at the product and store level only, not the individual customer level.

4. Exploratory Data Analysis

4.1 Overall Business Snapshot

Across all three stores, the business generated \$698,812.33 in total revenue from 214,470 units sold, at an average of \$4.69 per transaction. This spans 80 individual product SKUs grouped into 9 broader categories.

4.2 Revenue by Category

Revenue is not evenly spread across categories. Coffee and Tea alone account for two-thirds of all revenue, while the remaining seven categories share the rest.

Category	Revenue	Share of Total
Coffee	\$269,952.45	38.6%
Tea	\$196,405.95	28.1%
Bakery	\$82,315.64	11.8%
Drinking Chocolate	\$72,416.00	10.4%
Coffee Beans	\$40,085.25	5.7%
Branded	\$13,607.00	1.9%
Loose Tea	\$11,213.60	1.6%
Flavours	\$8,408.80	1.2%
Packaged Chocolate	\$4,407.64	0.6%

4.3 Top Products by Revenue

At the individual product level, Barista Espresso is the single largest revenue driver, followed by Brewed Chai Tea and Hot Chocolate.

Rank	Product	Revenue
1	Barista Espresso	\$91,406.20
2	Brewed Chai Tea	\$77,081.95
3	Hot Chocolate	\$72,416.00
4	Gourmet Brewed Coffee	\$70,034.60
5	Brewed Black Tea	\$47,932.00
6	Brewed Herbal Tea	\$47,539.50
7	Premium Brewed Coffee	\$38,781.15
8	Organic Brewed Coffee	\$37,746.50
9	Scone	\$36,866.12
10	Drip Coffee	\$31,984.00

4.4 Volume vs. Revenue - They Don't Always Match

Interestingly, Brewed Chai Tea sells in higher quantity (26,250 units) than Barista Espresso (24,943 units), even though Barista Espresso earns more total revenue. This gap shows why looking at volume alone can be misleading - a product can be more "popular" by unit count without being the strongest earner.

4.5 Revenue Concentration - the 80/20 Pattern

One of the clearest patterns in the data is how concentrated revenue is around a small set of products. The top 10 products alone account for 78.96% of total revenue, and it takes just 11 of the 29 product types to reach 80% of total revenue. This is a close match to the classic Pareto (80/20) principle, though slightly flatter than the textbook 80/20 split, meaning revenue is a little more spread out than a pure 80/20 rule would suggest - but still highly concentrated.

4.6 Underperforming Products

At the other end of the spectrum, a cluster of products consistently rank lowest in both units sold and revenue - Green Beans, Green Tea, House Blend Beans, Clothing, and Organic Chocolate among them. These products sell in the low hundreds of units for the entire year, compared to tens of thousands for the top performers.

5. Key Insights

- **Revenue is concentrated, not evenly spread.** 11 of 29 products (about 38%) generate 80% of total revenue - the business depends heavily on a small "hero" product set.
- **Coffee and Tea are the backbone categories.** Together they make up 66.7% of all revenue, while the remaining 7 categories split the rest.
- **Popularity and profitability aren't the same thing.** The top-selling product by volume (Brewed Chai Tea) is not the top earner (Barista Espresso) - decisions based on unit sales alone could be misleading.
- **A long tail of low-performing products exists.** A consistent group of products (largely beans, loose leaf items, and merchandise) underperform across both volume and revenue.

6. Recommendations

- **Protect and prioritize hero products.** Ensure the top 10-11 products (Barista Espresso, Brewed Chai Tea, Hot Chocolate, Gourmet Brewed Coffee, and similar) are always in stock and consistently high quality, since they carry the majority of revenue.
- **Review the underperforming product list for menu simplification.** Products like Green Beans, House Blend Beans, and Clothing contribute very little revenue relative to the menu space and inventory effort they require. These are candidates for bundling, discounting, or removal.
- **Use revenue-per-unit data for pricing decisions, not just volume.** Products with high revenue per unit (like Premium Beans at \$35.92/unit) show there is room to grow revenue through pricing and premium positioning, not just by selling more units.

7. Conclusion

This analysis shows that Afficionado Coffee Roasters' revenue is driven by a relatively small set of hero products rather than being spread evenly across the menu. Recognizing this pattern gives the business a clear, low-effort starting point for menu optimization: protect what already works, and review what doesn't. The full interactive breakdown of this analysis is available in the accompanying Power BI report and Streamlit dashboard.